

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 2: Polynomials

Previous year board questions, topic-wise: 6 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Zeroes and coefficients of a quadratic polynomial

1. The zeroes α and β of the polynomial $3x^2 + 6x + k$ satisfy $\alpha + \beta + \alpha\beta = -\frac{2}{3}$. Find k .
CBSE 2025, 1 mark
2. Find the zeroes of the polynomial $x^2 + \frac{4}{3}x - \frac{4}{3}$. CBSE 2025, 2 marks
3. The zeroes of $p(x) = 2x^2 - k\sqrt{2}x + 1$ add up to $\sqrt{2}$. Find k . (a) $\sqrt{2}$ (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$
CBSE 2024, 1 mark
4. Each zero of $x^2 + px + q$ is double a zero of $4x^2 - 5x - 6$. Find p . (a) $-\frac{5}{2}$ (b) $\frac{5}{2}$ (c) -5 (d) 10
CBSE 2024, 1 mark

2 Zeroes from the graph

5. The graphs of two quadratic polynomials are drawn on the same axes. One parabola opens upwards and the other opens downwards, and both cut the x -axis at the same two points. How many distinct zeroes do the two polynomials have between them? CBSE 2025, 1 mark
6. Assertion (A): a polynomial whose graph touches the x -axis at exactly one point cannot be quadratic. Reason (R): a polynomial of degree n , with $n > 1$, has at most n zeroes. Choose:
(a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A.
(c) A is true and R is false. (d) A is false and R is true. CBSE 2024, 1 mark

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